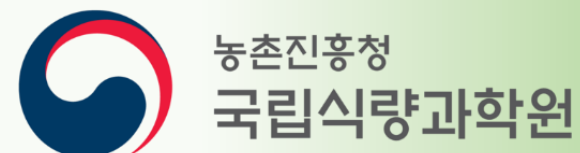


Effects of Organic Amendments Application on Soil Health, Microbial Diversity, and Growth Performance of Rice and Soybean Cropping Systems in the Honam Region

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Abstract

The field experiment was conducted to examine the effects of inorganic fertilizer with organic amendments. The results demonstrated that rice treated with NPK + Compost achieved the highest grain yield of 499.0 ± 20.1 kg ha⁻¹, which represents a remarkable 61% increase compared to the untreated control which yielded only 309.5 ± 17.4 kg ha⁻¹. Interestingly, NPK fertilization alone produced comparable results with 498.3 ± 4.2 kg ha⁻¹, suggesting that both approaches substantially enhanced productivity. For soybean production, NPK + compost yielded 206.4 ± 10.2 kg ha⁻¹, representing a 4.6% improvement over the control. The compost amendment markedly improved soil chemical properties, most notably increasing available phosphorus in the topsoil by 82%, rising from 37-38.5 mg/kg in control and NPK-only treatments to 68.4 mg/kg in the compost treatment. Organic matter content also increased significantly, reaching 16% with compost application. Soil physical analyses revealed that NPK + Compost preserved soil porosity better than NPK alone, with porosity maintained at 39.8% compared to just 35.9% under NPK-only treatment, thereby reducing the risk of soil compaction. Among the microbial community analysis the Chao1 index varied significantly among treatments, and principal coordinates analysis showed partial separation of microbial communities according to fertilizer regime. At the phylum level, communities were dominated by *Proteobacteria*, *Acidobacteria*, and *Chloroflexi*. At the genus level, beneficial microorganisms including *Bradyrhizobium*, *Bacillus*, *Sphingomonas*, and *Streptomyces* showed increased relative abundance under compost application. These comprehensive findings demonstrate that integrated application of inorganic fertilizers combined with organic amendments substantially enhances nutrient availability, preserves soil physical structure, maintains microbial ecosystem integrity, and improves crop productivity in both rice and soybean systems.

[Introduction]: Long-term fertilization and the use of proper fertilizer sources are essential to boost nutrient efficiency [1]. The application of compost in agroecosystems is recognized to enhance soil fertility and increase agricultural productivity while also improving soil health [2]. Integrated nutrient management combining inorganic fertilizers with organic inputs further improves carbon sequestration and soil physical health [3]. This study aims to propose long-term soil and crop growth management strategies for paddy and upland fields in the Honam region under different fertilizer regimes. Based on these baseline data, future work will incorporate paddy-to-upland field conversion scenarios to develop a comprehensive database on soil and crop growth changes.

[Materials and Methods]: Field experiments were established in 2025. Rice plots received three treatments: Control (no amendment), NPK, and NPK + Compost. Soybean cropping received identical treatments. Agronomic traits (grain yield, tillers, panicles, pod and seed numbers, stem length, stem diameter) were recorded at harvest. Soil samples were collected from topsoil (0–20 cm) and subsoil (20–40 cm) before transplanting and after harvest for analysis of bulk density, porosity, water content, pH, organic matter, available phosphorus, total nitrogen, and exchangeable cations. Soil microbial communities were characterized using high-throughput sequencing; alpha diversity (Chao1), beta diversity (principal coordinates analysis based on Jensen–Shannon distance), and taxonomic composition at the phylum and genus levels were assessed.

[Results and Discussion]: For rice, the NPK + Compost treatment produced the highest grain yield (499.0 ± 20.1 kg ha⁻¹), representing a 61% increase relative to the control (309.5 ± 17.4 kg ha⁻¹), followed closely by NPK alone (498.3 ± 4.2 kg ha⁻¹). Most agronomic traits improved with fertilization. Soil physical changes were modest; bulk density slightly increased and porosity decreased under fertilized treatments. Chemical properties remained favorable, with stable pH and minimal treatment differences in organic matter and available phosphorus after the first season. Microbial alpha diversity differed among treatments, and principal coordinates analysis revealed partial separation of communities according to fertilizer regime. Phylum-level composition was dominated by *Proteobacteria*, *Acidobacteria*, and *Chloroflexi*. For soybean, NPK + Compost produced the highest grain yield (206.4 ± kg ha⁻¹), representing a 4.6 % increase over the control (197.3 ± 15.6 kg ha⁻¹), compared to NPK alone (193.6 ± 24.1 kg ha⁻¹). NPK + Compost markedly increased available phosphorus in the topsoil by 82% (68.4 mg kg⁻¹ versus 37–38.5 mg kg⁻¹), with the highest organic matter content at 16%. In the subsoil, NPK alone elevated bulk density to 1.70 g/cm³ and reduced porosity to 35.9%, indicating a compaction risk, whereas NPK + Compost maintained higher porosity (39.8%). Microbial analysis revealed partial separation of communities, with the compost treatment showing distinct clustering. Taxa including *Bradyrhizobium*, *Bacillus*, *Sphingomonas*, and *Streptomyces* showed increased relative abundance under compost application. These baseline results demonstrate that the integrated application of inorganic fertilizers and organic amendments improves nutrient availability, preserves soil physical structure, supports microbial community integrity, and enhances crop productivity in both rice and soybean systems.

1. Agronomic Performance and Grain Yield

Table 1. Rice Continuous

	Rice Continuous Plot									
	Culm + panicle length	Number of Tiller	Number of Panicle	Stem length	No of grains perpanicle	SPAD Value	1000 grains Weight	Grain Filling Ratio	Yield kg ha ⁻¹	
Control	82.83 ± 1.62	9.08 ± 0.26	8.74 ± 1	60.26 ± 1.3	266.77 ± 25.44	19.74 ± 1.12	27.34 ± 0.42	92.39 ± 0.86	309.46 ± 17.36	
NPK	93.64 ± 0.27	11.65 ± 1.48	11.44 ± 8.74 ± 1.05	69.83 ± 0.53	297.78 ± 6.77	23.27 ± 0.94	28.3 ± 0.54	92.87 ± 0.55	498.34 ± 4.16	
NPK+Compost	94.74 ± 0.94	11.81 ± 1.64	13.37 ± 1.97	71.19 ± 0.5	290.46 ± 6.32	24.01 ± 0.81	28.9 ± 0.23	93.23 ± 0.78	498.98 ± 20.09	

NPK + Compost produced the highest grain yield (499.0 ± 20.1 kg ha⁻¹), a 61% increase over the unfertilized control (309.5 ± 17.4 kg ha⁻¹). NPK alone yielded nearly the same (498.3 ± 4.2 kg ha⁻¹). Most other agronomic traits (tiller number, panicle number, stem length, SPAD value, 1000-grain weight, grain-filling ratio) also improved under both fertilized treatments relative to the control.

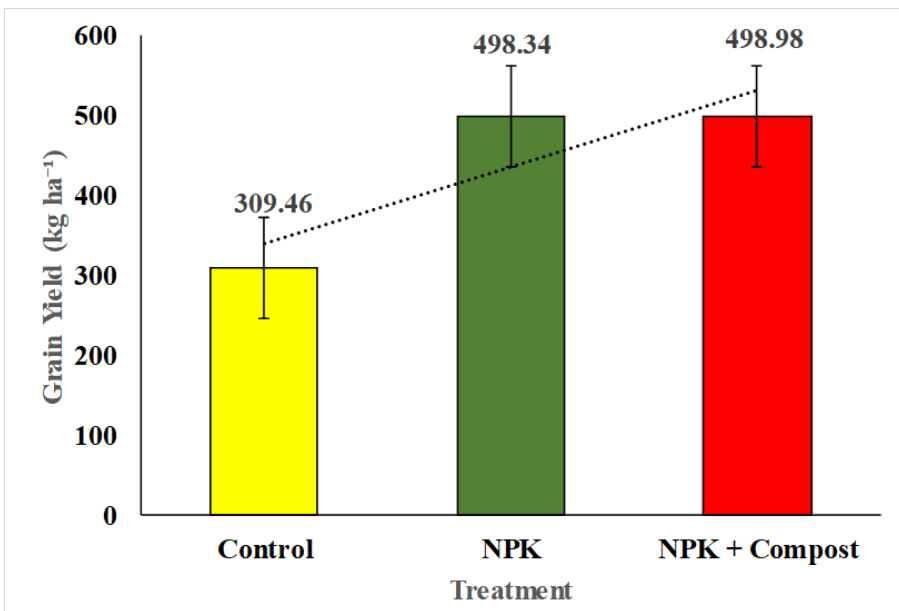


Figure 1. Rice grain yield under fertilized treatments

Table 2. Soybean Continuous

	Soybean Plot							
	Stem Length	Stem Diameter	Number of Pods	Number of seeds	Number of Nodes	Number of Branches	1000 Seeds Weight	Yield kg ha ⁻¹
Control	58.8 ± 1.42	8.62 ± 0.63	52.85 ± 4.07	71.11 ± 7.57	16.86 ± 0.21	3.38 ± 0.44	27.34 ± 0.18	197.34 ± 15.57
NPK	58.84 ± 0.51	8.53 ± 0.42	54.12 ± 4.17	72.27 ± 9.89	16.67 ± 0.4	3.42 ± 0.13	27.69 ± 0.92	193.59 ± 24.07
NPK+Compost	63.57 ± 1.43	8.7 ± 0.29	53.49 ± 4.61	74.87 ± 7.45	17.09 ± 0.14	3.48 ± 0.33	27.27 ± 0.88	206.39 ± 10.19

NPK + Compost again gave the highest grain yield (206.4 ± 10.2 kg ha⁻¹), a modest 4.6% increase over the control (197.3 ± 15.6 kg ha⁻¹). NPK alone produced a slightly lower yield (193.6 ± 24.1 kg ha⁻¹). Stem length, pod and seed numbers, and 1000-seed weight also responded favorably to the combined treatment (Table 2 and Figure 2).

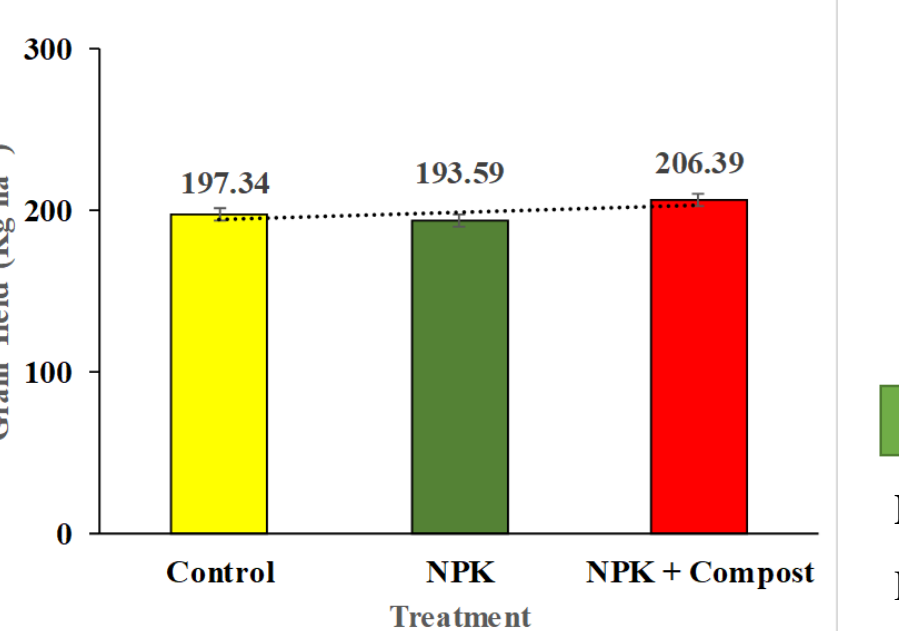


Figure 2. Soybean grain yield under fertilized treatments

2. Soil physical properties of Rice and Soybean Continuous Plot

Table 3. Rice Continuous

	Rice Continuous Plot			
	Soil Physical properties	Control	NPK	NPK + Compost
Before Transplanting	Bulk density g/cm ³	1.33 ± 0	1.36 ± 0.01	1.39 ± 0.01
	Soil Porosity	49.94 ± 0	48.58 ± 0.01	47.56 ± 0
	Gravimetric Water Content (%)	36.81 ± 0.01	34.94 ± 0	33.61 ± 0.01
	Sub Soil			
After Harvesting	Bulk density g/cm ³	1.42 ± 0.01	1.55 ± 0.01	1.52 ± 0.01
	Soil Porosity	46.44 ± 0.01	41.57 ± 0	42.81 ± 2.01
	Gravimetric Water Content (%)	31.93 ± 0.01	26.98 ± 0	28.07 ± 0
	Sub Soil			
Before Transplanting	Bulk density g/cm ³	1.48 ± 0.05	1.47 ± 0.06	1.47 ± 0.12
	Soil Porosity	44.26 ± 4.35	44.63 ± 1.57	44.5 ± 2.24
	Gravimetric Water Content (%)	28.77 ± 2.65	29.36 ± 2.49	28.11 ± 4.68
	Sub Soil			
After Harvesting	Bulk density g/cm ³	1.61 ± 0.06	1.59 ± 0.06	1.62 ± 0.06
	Soil Porosity	39.08 ± 2.2	40.08 ± 1.61	38.77 ± 0
	Gravimetric Water Content (%)	24.57 ± 2.01	25.69 ± 2.11	25.16 ± 3
	Sub Soil			

In rice continuous plot changes were modest. Bulk density tended to increase slightly and porosity decreased under the fertilized treatments, especially in the subsoil, but differences remained small after one season. On the other hand the subsoil, of soybean plot showed that NPK alone raised bulk density to 1.70 g cm⁻³ and reduced porosity to 35.9%, indicating a risk of compaction. In contrast, NPK + Compost maintained higher porosity (39.8%), suggesting that the organic amendment helped preserve soil structure.

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3. Soil Chemical Properties

Tables 5. Rice Continuous

	Rice Continuous Plot			
	Top Soil 0-20 (cm)	Control	NPK	NPK + Compost
Before Transplantation	Chemical Properties			
	pH	6.76 ± 0.32	6.64 ± 0.18	6.86 ± 0.17
	EC	0.22 ± 0.03	0.21 ± 0.02	0.22 ± 0.01
	Organic Matter	14.31 ± 0.92	14.05 ± 1.08	13.13 ± 0.83
	Av.Phosphorus	26.68 ± 3.67	25.47 ± 4.22	24.98 ± 3.89
	TN	0.09 ± 0.01	0.09 ± 0.01	0.09 ± 0.01
	Ca	6.74 ± 0.35	6.71 ± 0.40	6.82 ± 0.39
	K	0.23 ± 0.03	0.23 ± 0.02	0.22 ± 0.03
	Mg	2.47 ± 0.20	2.53 ± 0.20	2.64 ± 0.21
	Na	0.18 ± 0.01	0.19 ± 0.01	0.2 ± 0.01
	Sub Soil 20-40 (cm)			
	pH	7.25 ± 0.35	7.23 ± 0.13	7.17 ± 0.11
After Harvesting	Chemical Properties			
	pH	0.22 ± 0.02	0.22 ± 0.03	0.21 ± 0.04
	EC	9.72 ± 2.05	10.39 ± 1.23	9.44 ± 1.54
	Organic Matter	19.71 ± 8.26	21.21 ± 8.5	23.4 ± 6.83
	Av.Phosphorus	19.71 ± 8.26	21.21 ± 8.5	23.4 ± 6.83
	TN	0.07 ± 0.02	0.07 ± 0.01	0.06 ± 0.01
	Ca	7.24 ± 0.77	7.3 ± 0.61	6.93 ± 0.58
	K	0.19 ± 0.02	0.19 ± 0.03	0.19 ± 0.01
	Mg	2.98 ± 0.37	3.02 ± 0.35	3.05 ± 0.43
	Na	0.21 ± 0.02	0.21 ± 0.02	0.22 ± 0.03
	Sub Soil 20-40 (cm)			
	pH	6.82 ± 0.16	6.80 ± 0.11	6.87 ± 0.28
	EC	0.44 ± 0.09	0.42 ± 0.09	0.4 ± 0.06
	Organic Matter	11.18 ± 3.33	10.52 ± 1.79	9.87 ± 3.5
	Av.Phosphorus	27.27 ± 7.45	29.14 ± 8.43	22.69 ± 9.87
	TN	0.07 ± 0.01	0.06 ± 0.01	0.06 ± 0.02
	Ca	6.91 ± 0.33	6.73 ± 0.44	7.02 ± 0.85
	K	0.23 ± 0.02	0.23 ± 0.02	0.21 ± 0.03
	Mg	2.83 ± 0.33	2.55 ± 0.31	3.22 ± 0.98
	Na	0.3 ± 0.05	0.29 ± 0.03	0.3 ± 0.05

In rice continuous plot soil pH stayed stable across treatments. Organic-matter content and available phosphorus showed only minimal treatment differences after the first season. Overall chemical fertility remained favorable under all regimes (Table 5 and 6).

Table 7. Soybean Continuous

	Soybean Continuous Plot			
	Top Soil 0-20 (cm)	Control	NPK	NPK + Compost
Before Transplantation	Chemical Properties			
	pH	6.76 ± 0.09	6.75 ± 0.1	6.78 ± 0.14
	EC	0.52 ± 0.14	0.46 ± 0.07	0.45 ± 0.06
	Organic Matter	14.6 ± 2.18	13.48 ± 0.97	13.95 ± 1.7
	Av.Phosphorus	35.08 ± 11.31	34.54 ± 6.86	33.62 ± 8.99
	TN	0.09 ± 0.01	0.08 ± 0.01	0.08 ± 0.02
	Ca	7.03 ± 0.31	7.03 ± 0.26	6.97 ± 0.31
	K	0.27 ± 0.03	0.27 ± 0.03	0.27 ± 0.03
	Mg	2.49 ± 0.15	2.46 ± 0.13	2.55 ± 0.26
	Na	0.31 ± 0.04	0.31 ± 0.03	0.3 ± 0.03
	Sub Soil 20-40 (cm)			
	pH	6.82 ± 0.16	6.80 ± 0.11	6.87 ± 0.28
After Harvesting	Chemical Properties			
	pH	0.44 ± 0.09	0.42 ± 0.09	0.4 ± 0.06
	EC	11.18 ± 3.33	10.52 ± 1.79	9.87 ± 3.5
	Organic Matter	27.27 ± 7.45	29.14 ± 8.43	22.69 ± 9.87
	Av.Phosphorus	0.07 ± 0.01	0.06 ± 0.01	0.06 ± 0.02
	TN	6.91 ± 0.33	6.73 ± 0.44	7.02 ± 0.85
	Ca	0.23 ± 0.02	0.23 ± 0.02	0.21 ± 0.03
	K	2.83 ± 0.33	2.55 ± 0.31	3.22 ± 0.98
	Mg	0.3 ± 0.05	0.29 ± 0.03	0.3 ± 0.05
	Na	0.31 ± 0.03	0.31 ± 0.06	0.33 ± 0.03
	Sub Soil 20-40 (cm)			
	pH	7.17 ± 0.11	7.18 ± 0.19	7.13 ± 0.14
	EC	0.39 ± 0.08	0.37 ± 0.07	0.4 ± 0.09
	Organic Matter	8.96 ± 1.71	9.13 ± 1.02	9.34 ± 1.9
	Av.Phosphorus	19.85 ± 10.76	22.93 ± 4.04	20.73 ± 13.44
	TN	0.06 ± 0.01	0.05 ± 0.01	0.05 ± 0.01
	Ca	6.56 ± 0.53	6.76 ± 0.66	6.38 ± 0.53
	K	0.21 ± 0.01	0.23 ± 0.03	0.21 ± 0.02
	Mg	2.86 ± 0.41	2.65 ± 0.3	2.99 ± 0.53
	Na	0.31 ± 0.03	0.31 ± 0.06	0.33 ± 0.03

In soybean plot the most striking change was in the topsoil: available phosphorus rose by approximately 82% under NPK + Compost (68.4 mg kg⁻¹ versus 37–38.5 mg kg⁻¹ in the other treatments). Organic-matter content also reached its highest value (~16%) with the compost-amended treatment (Table 7 and 8).

4. Rice Continuous Plot Microbial Diversity

Phylum Level

The community was dominated by *Proteobacteria* (largest share, ~30–35%), followed by *Chloroflexi*, *Actinobacteria*, and *Acidobacteria*. Smaller contributions came from *Planctomycetes*, *Nitrospirae*, *Firmicutes*, *Bacteroidetes*, and *Verrucomicrobia*. Composition was quite similar across Control, NPK, and NPK + Compost, with only minor shifts.

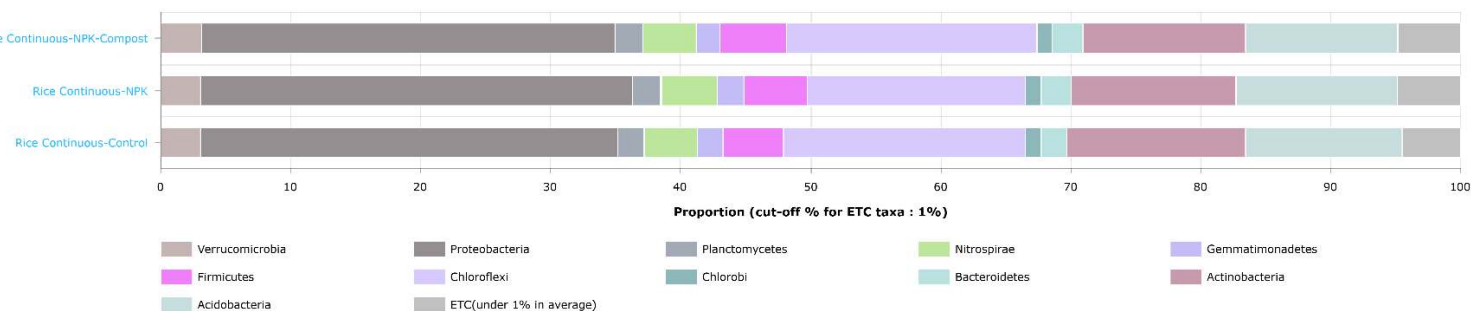


Figure 3. Microbial diversity at phylum level

Chao1 (alpha diversity)

Fertilized treatments (NPK and NPK + Compost) showed higher richness than the Control. Median Chao1 was highest under NPK (~4,800), followed by NPK + Compost (~4,550), and lowest in Control (~4,450). Variation was larger in the Control.

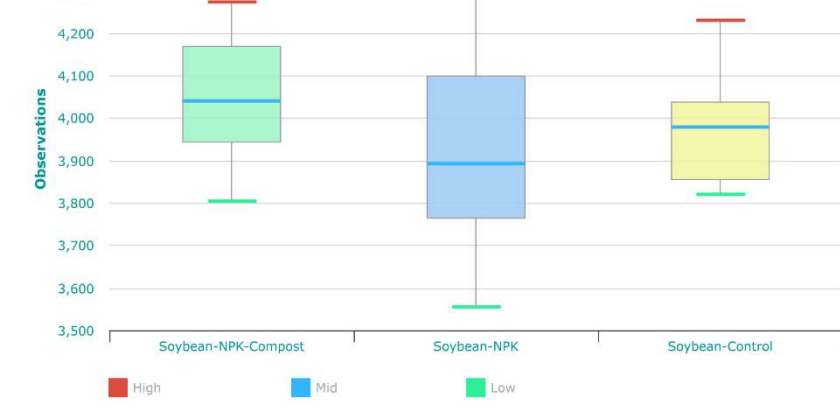


Figure 5. Alpha diversity under different treatment groups

5. Soybean Continuous Plot

Phylum level

Proteobacteria dominated (~35%), followed by *Actinobacteria*, *Acidobacteria*, *Chloroflexi*, and *Gemmatimonadetes*. *Firmicutes*, *Bacteroidetes*, *Planctomycetes*, and *Verrucomicrobia* made smaller contributions. Composition was relatively stable across treatments, with slight enrichment of some groups under compost.

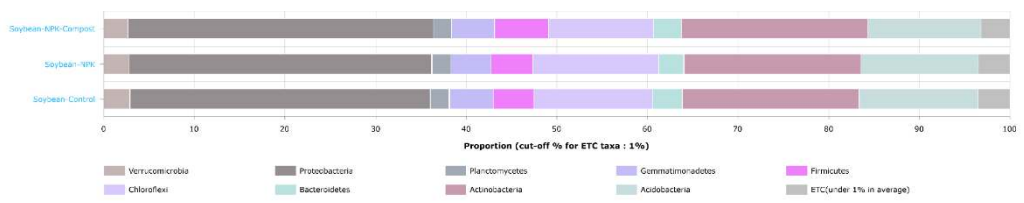


Figure 7. Microbial diversity at phylum level under fertilizer amendment

PCoA (beta diversity)

Clearer partial separation was observed. Compost samples (green) tended to cluster toward the positive side of the first axis (PC1 = 19.4%), while Control (yellow) and NPK (blue) were more mixed on the left. PC2 explained ~15.0%. The compost treatment formed a more distinct group, indicating that organic amendment shifted the overall community structure more strongly than NPK alone.

Tables 6. Rice Continuous

	Rice Continuous Plot			
	Top Soil 0-20 (cm)	Control	NPK	NPK + Compost
Before Transplantation	Chemical Properties			
	pH	7.17 ± 0.33	7.09 ± 0.18	7.33 ± 0.49
	EC	0.2 ± 0.02	0.23 ± 0.01	0.24 ± 0.02
	Organic Matter	13.14 ± 0.69	13.61 ± 0.87	14.35 ± 1.1
	Av.Phosphorus	19.62 ± 3.7	18.8 ± 1.08	20.29 ± 2.68
	TN	0.1 ± 0.05	0.08 ± 0.01	0.08 ± 0.01
	Ca	5.62 ± 0.42	5.89 ± 0.35	5.82 ± 0.34
	K	0.23 ± 0.02	0.23 ± 0.02	0.26 ± 0.03
	Mg	2.12 ± 0.22	2.29 ± 0.22	2.12 ± 0.2
	Na	0.18 ± 0.02	0.19 ± 0.01	0.19 ± 0.01
	Sub Soil 20-40 (cm)			
	pH	7.4 ± 0.16	7.72 ± 0.56	7.51 ± 0.2